

β -OPSD: Deriving with Policy Optimization, Training with Self-Distillation

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On-policy self-distillation (OPSD) is a promising approach to improve reasoning language models, but it remains brittle in practice: making it work reliably often requires substantial engineering effort. We identify a structural source of this difficulty: vanilla OPSD is precisely the $\beta = 1$ member of a broader policy-optimization family, where β weights the KL penalty anchoring the student to a reference policy. This equivalence turns β from an implicit value fixed at one into a controllable regularization parameter, yielding a more general formulation that trades off proximity to a reference policy against privileged teacher guidance. We introduce **β -OPSD** and derive its optimal policy as a geometric interpolation between the reference policy and the privileged teacher. Directly optimizing this objective with reinforcement learning, however, would be costly and high-variance. Rather than optimize the RL objective directly, we turn its closed-form solution into a distillation target. Each value of β selects a target along the reference-to-teacher path, which we implement efficiently by mixing their token-level logits. In this way, inexpensive distillation approximates the solution of expensive policy optimization. Return-to-go credit assignment further aligns token updates with the sequence-level objective while retaining the simplicity of OPSD. Experiments on mathematical reasoning benchmarks show that β -OPSD consistently outperforms vanilla OPSD, improving optimization stability and downstream reasoning performance. Our results provide a principled route from self-distillation to policy optimization and back—without sacrificing the efficiency that makes OPSD practical.

β -OPSD: derive the target with policy optimization, train it with self-distillation

A controllable path from the reference policy to privileged teacher guidance

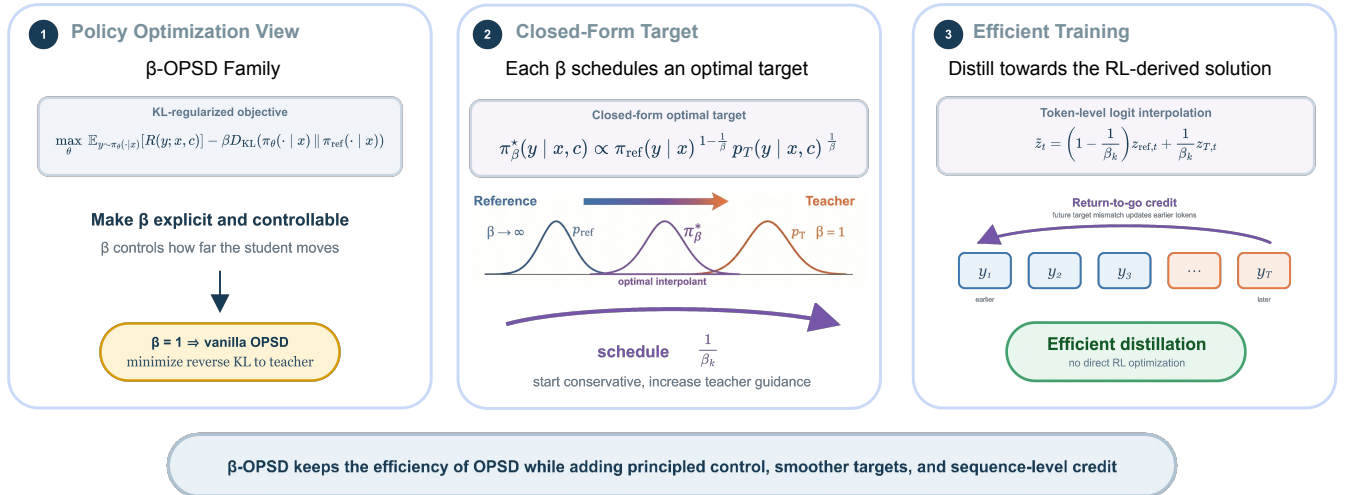


Figure 1: Overview of β -OPSD. Vanilla OPSD is the $\beta = 1$ case of a broader KL-regularized objective, whose optimal policy is a geometric interpolant between the reference policy and the privileged teacher. β -OPSD turns this optimal policy into a practical distillation target through scheduled logit interpolation, while retaining efficient token-level training with return-to-go credit assignment.

* Equal contribution and joint project leadership.

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1. Introduction

Reasoning performance of large language models has advanced rapidly through reinforcement learning and self-improvement techniques (Shao et al., 2024, Ouyang et al., 2022, Guo et al., 2025). Among these approaches, on-policy self-distillation (OPSD) is a promising way to improve reasoning: the student learns from its own sampled trajectories while a teacher supplies supervision using privileged information available only during training, such as reference solutions, verified reasoning traces, or external feedback (Zhao et al., 2026, Shenfeld et al., 2026, Hübottter et al., 2026, Li et al., 2026, Jang et al., 2026, Yang et al., 2026). Unlike offline distillation, OPSD lets the teacher respond to the student’s actual intermediate states. Yet this promise has not produced a reliably effective recipe; in practice, OPSD can be brittle and often demands substantial engineering.

Most existing OPSD methods begin from direct teacher imitation. Given a student-generated trajectory, the privileged teacher provides token-level predictions, and the student minimizes a reverse Kullback–Leibler (KL) divergence to those predictions (Zhao et al., 2026, Jang et al., 2026, Li et al., 2026, Yang et al., 2026). This formulation fixes the teacher as the unique optimization target and provides no explicit control over how aggressively the student should move away from its current or initial policy. Is direct teacher matching truly the only principled OPSD objective?

Our key observation is that it is not. Vanilla OPSD is precisely the $\beta = 1$ member of a broader KL-regularized policy-optimization family, where β weights the KL penalty that anchors the learned policy to a reference policy. This equivalence turns β from an implicit value fixed at one into an explicit control over the strength of teacher guidance. It also changes the interpretation of OPSD: rather than a single teacher-imitation objective, it is one point in a continuum of policy-optimization problems that trade off improvement toward the privileged teacher against proximity to the reference policy.

This perspective leads to **β -OPSD**, a generalized family of on-policy self-distillation objectives parameterized by β . Vanilla OPSD is recovered exactly at $\beta = 1$, while larger values yield more conservative updates that remain closer to the reference policy. The family therefore exposes a degree of freedom that vanilla OPSD leaves fixed and offers a principled mechanism for addressing its brittleness: instead of moving abruptly toward the teacher, the target can progress smoothly from the reference policy toward privileged teacher guidance over the course of training.

The policy-optimization view also tells us what these targets should be. We derive the optimal policy of β -OPSD in closed form and show that it is a geometric interpolation between the reference policy and the privileged teacher. Directly optimizing the resulting RL objective, however, would incur costly rollouts, advantage estimation, and high-variance gradients; the exact sequence-level optimum also contains an intractable normalizer. We therefore use policy optimization for the derivation and return to distillation for training. Each value of β defines an RL-derived target along the reference-to-teacher path, which we realize efficiently through token-level logit interpolation. Scheduling β then gives a smooth curriculum of targets while preserving the computational structure of OPSD. In short, inexpensive distillation approximates the solution of expensive policy optimization.

Finally, matching the right target still requires assigning credit across the generated trajectory. Standard token-local OPSD updates are myopic: an early token changes later prefixes and therefore affects future student–target mismatch. A policy-gradient derivation of the sequence-level objective yields return-to-go credit assignment, which propagates this future mismatch back to earlier tokens. Combined with the interpolated target, this gives a practical algorithm that retains token-level distillation while more faithfully optimizing the underlying sequence-level objective.

We evaluate β -OPSD on mathematical reasoning benchmarks using open-source reasoning language models ranging from 1.7B to 8B parameters. Across multiple benchmarks, β -OPSD consistently improves optimization stability and reasoning performance over vanilla OPSD. On Qwen3-1.7B (Yang et al., 2025), it improves avg@12 by up to 9.16 percentage points over vanilla OPSD across AIME 2024 (Zhang and Math-AI, 2024), AIME 2025 (Zhang and Math-AI, 2025), and HMMT 2025 (Dekoninck et al., 2026). Ablations further isolate the benefits of both the RL-derived interpolation target and sequence-level return-to-go credit assignment.

Our contributions are summarized as follows:

- We show that vanilla OPSD is precisely the $\beta = 1$ member of a broader KL-regularized policy-optimization family, making its implicit regularization choice explicit and controllable.
- We introduce **β -OPSD** and derive its closed-form optimal policy as a geometric interpolation between the reference policy and the privileged teacher.
- We turn the RL-derived solution back into an efficient distillation algorithm using scheduled logit interpolation and return-to-go credit assignment.
- We demonstrate consistent improvements over vanilla OPSD on mathematical reasoning benchmarks and validate the contribution of both practical components.

2. Method

This section develops β -OPSD following the paper’s central principle: derive the target with policy optimization, then train it with self-distillation. We first show that vanilla OPSD is precisely the $\beta = 1$ member of a broader KL-regularized policy-optimization family, making the strength of reference regularization explicit and controllable. We then derive the family’s optimal policy as a geometric interpolation between the reference policy and the privileged teacher. Because direct RL optimization is costly and the exact sequence-level target is intractable, we return to efficient distillation by realizing the target through scheduled token-level logit interpolation. Finally, we derive return-to-go credit assignment to align the practical token-level update with the underlying sequence-level objective. Appendix A provides a consolidated reference for the notation used throughout the paper.

2.1. Vanilla OPSD as $\beta = 1$ Policy Optimization

Let $\pi_\theta(y \mid x)$ denote the student policy, $p_T(y \mid x, c)$ the privileged teacher, and $\pi_{\text{ref}}(y \mid x)$ a reference policy, such as the initial student or a stop-gradient copy of the current student. We start from the standard KL-regularized RL objective

$$\max_{\theta} \mathbb{E}_{y \sim \pi_\theta(\cdot \mid x)} [R(y; x, c)] - \beta D_{\text{KL}}(\pi_\theta(\cdot \mid x) \parallel \pi_{\text{ref}}(\cdot \mid x)), \quad (1)$$

where $\beta > 0$ controls how strongly the learned policy is constrained to remain close to the reference.

To embed standard on-policy self-distillation within this policy-optimization family, we choose the reward to be the teacher-to-reference log-ratio:

$$R(y; x, c) = \log \frac{p_T(y \mid x, c)}{\pi_{\text{ref}}(y \mid x)}. \quad (2)$$

This reward is high when a trajectory is more likely under the privileged teacher than under the reference

policy. Substituting Eq. 2 into Eq. 1 gives the family

$$\mathcal{J}_\beta(\theta) = \mathbb{E}_{y \sim \pi_\theta(\cdot | x)} \left[\log \frac{p_T(y | x, c)}{\pi_{\text{ref}}(y | x)} \right] - \beta D_{\text{KL}}(\pi_\theta(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)). \quad (3)$$

We call this objective β -OPSD. It generalizes standard on-policy self-distillation by treating β as a variable regularization parameter rather than implicitly fixing it at one. The log-ratio reward favors trajectories preferred by the privileged teacher over the reference policy, while the KL term penalizes movement away from the reference; β controls the trade-off between these two forces.

Proposition 2.1 (OPSD is the $\beta = 1$ case). *When $\beta = 1$, maximizing Eq. 3 is equivalent to minimizing the standard sequence-level reverse KL,*

$$D_{\text{KL}}(\pi_\theta(\cdot | x) \| p_T(\cdot | x, c)). \quad (4)$$

At $\beta = 1$, the reference-policy terms in the log-ratio reward and KL penalty cancel, leaving exactly the negative reverse KL from the student to the privileged teacher. Thus, standard OPSD is not separate from policy optimization; it is one member of this KL-regularized family. For $\beta > 1$, the remaining reference penalty makes the update more conservative, while decreasing β toward 1 moves the objective more aggressively toward the teacher. The proof is given in Appendix B.1.

2.2. Variable β Defines a Path of Optimal Targets

Once vanilla OPSD is recognized as the $\beta = 1$ case, the next question is what target each value of β defines. For any fixed regularization coefficient, the optimal policy of β -OPSD can be written in closed form.

Proposition 2.2 (Optimal policy of β -OPSD). *Assuming compatible support, the maximizer of Eq. 3 is*

$$\pi_\beta^*(y | x, c) = \frac{\pi_{\text{ref}}(y | x)^{1-\frac{1}{\beta}} p_T(y | x, c)^{\frac{1}{\beta}}}{Z_\beta(x, c)}, \quad (5)$$

where $Z_\beta(x, c)$ normalizes over full trajectories.

The proof is in Appendix B.2. The solution is a geometric interpolation between the reference policy and the privileged teacher. For $\beta \geq 1$, the teacher weight $\frac{1}{\beta}$ lies in $[0, 1]$: $\beta = 1$ recovers the teacher endpoint, while $\beta \rightarrow \infty$ approaches the reference endpoint. Thus, the teacher is no longer the unique target. Each value of β selects a different optimal policy along a principled reference-to-teacher path.

Scheduling the interpolation. The theory above defines a family of targets for fixed β . In practice, we exploit this new degree of freedom by varying the coefficient over training. We write the coefficient at step k as β_k ; its inverse directly determines the teacher weight. We require

$$\frac{1}{\beta_k} \in [0, 1], \quad \text{equivalently } \beta_k \geq 1, \quad (6)$$

so that the target remains on the path between the reference and the teacher. A larger $\frac{1}{\beta_k}$ gives a more teacher-like target, while a smaller $\frac{1}{\beta_k}$ keeps the target closer to the reference policy.

In practice, we use a bounded linear schedule for the teacher weight:

$$\frac{1}{\beta_k} = w_{\text{start}} + (w_{\text{end}} - w_{\text{start}}) \frac{k}{K-1}, \quad k = 0, \dots, K-1. \quad (7)$$

Equivalently,

$$\beta_k = \left[w_{\text{start}} + (w_{\text{end}} - w_{\text{start}}) \frac{k}{K-1} \right]^{-1}. \quad (8)$$

When $w_{\text{start}} < w_{\text{end}}$, early targets remain closer to the reference policy, while later targets incorporate stronger teacher guidance. This smooth curriculum avoids an abrupt projection onto the privileged teacher and provides a direct mechanism for reducing the brittleness of vanilla OPSD.

Local logit realization. Although Eq. 5 gives the optimal trajectory-level distribution, it is not directly usable for language models because the normalizer $Z_\beta(x, c)$ ranges over all possible completions. We therefore approximate the optimal policy locally at each decoding prefix.

Let

$$h_t = (x, y_{<t}) \quad (9)$$

be the prefix at token t . Let $z_{\text{ref}}(\cdot | h_t)$ denote the reference or student-side logits, and let $z_T(\cdot | h_t, c)$ denote the privileged teacher logits. Given the scheduled coefficient β_k , we define the local logit-interpolant target as

$$\tilde{p}_{\beta_k}(\cdot | h_t, c) = \text{softmax} \left(\left(1 - \frac{1}{\beta_k} \right) z_{\text{ref}}(\cdot | h_t) + \frac{1}{\beta_k} z_T(\cdot | h_t, c) \right). \quad (10)$$

At each prefix, this distribution is exactly the normalized geometric interpolation of the local reference and teacher distributions: multiplying probabilities geometrically is equivalent, up to normalization, to linearly interpolating their logits.

The resulting autoregressive interpolant target is

$$\tilde{p}_{\beta_k}(y | x, c) = \prod_{t=1}^T \tilde{p}_{\beta_k}(y_t | x, y_{<t}, c). \quad (11)$$

Because normalization is applied separately at each prefix, the resulting autoregressive distribution is a tractable local approximation to the sequence-level optimum in Eq. 5. This is the target used by the practical distillation objective below.

2.3. From Policy Optimization Back to Distillation

The β -OPSD objective in Eq. 3 can in principle be optimized with KL-regularized RL algorithms. Doing so, however, would introduce additional RL machinery, such as reward-weighted policy updates and advantage estimation, together with higher-variance gradients. The analysis above provides a cheaper route: the policy objective specifies an optimal sequence-level target, and Eq. 11 gives its tractable local approximation.

This is the main “back to distillation” step. Rather than directly maximizing the RL objective, at training step k we train the student to match the RL-derived interpolant target $\tilde{p}_{\beta_k}$ by minimizing

$$\mathcal{L}_{\beta_k}(\theta) = D_{\text{KL}}(\pi_\theta(\cdot | x) \| \tilde{p}_{\beta_k}(\cdot | x, c)). \quad (12)$$

Equivalently,

$$\mathcal{L}_{\beta_k}(\theta) = \mathbb{E}_{y \sim \pi_\theta(\cdot | x)} \left[\log \frac{\pi_\theta(y | x)}{\tilde{p}_{\beta_k}(y | x, c)} \right]. \quad (13)$$

Using the autoregressive factorization in Eq. 11, this becomes

$$\mathcal{L}_{\beta_k}(\theta) = \mathbb{E}_{y \sim \pi_\theta(\cdot | x)} \left[\sum_{t=1}^T \rho_t^{\beta_k}(y) \right], \quad (14)$$

where

$$\rho_t^{\beta_k}(y) = \log \pi_\theta(y_t | x, y_{<t}) - \log \tilde{p}_{\beta_k}(y_t | x, y_{<t}, c). \quad (15)$$

This objective has the same form as vanilla OPSD, except that the fixed teacher p_T is replaced by the scheduled interpolant target $\tilde{p}_{\beta_k}$. When $\beta_k = 1$, the interpolant target reduces to the privileged teacher, and Eq. 12 recovers standard OPSD. For larger β_k , the target remains closer to the reference policy, yielding a more conservative distillation objective.

The interpolant target is efficient to compute. Once the reference/student logits and teacher logits are available, constructing $\tilde{p}_{\beta_k}$ only requires an elementwise weighted sum of the two logit vectors followed by a softmax, as in Eq. 10. Importantly, this avoids the intractable sequence-level normalizer $Z_\beta(x, c)$ in Eq. 5. Thus, policy optimization specifies where the student should move, while token-level distillation provides an inexpensive way to get there.

2.4. Practical β -OPSD with Return-to-Go Credit

We now describe the practical training loss. At each update, we sample trajectories on-policy from the current student:

$$y \sim \pi_\theta(\cdot | x). \quad (16)$$

Along the sampled trajectory, we evaluate the token-level mismatch between the student and the interpolant target:

$$\rho_t^{\beta_k}(y) = \log \pi_\theta(y_t | x, y_{<t}) - \log \tilde{p}_{\beta_k}(y_t | x, y_{<t}, c). \quad (17)$$

The scalar estimator $\sum_t \rho_t^{\beta_k}(y)$ is an unbiased Monte Carlo estimate of the sequence-level reverse KL in Eq. 12. However, using each token’s instantaneous log-ratio as its training weight gives a myopic gradient: it ignores how an earlier token affects future prefixes and future target mismatch.

Therefore, β -OPSD uses a return-to-go weight:

$$G_{t,\gamma}^{\beta_k}(y) = \sum_{s=t}^T \gamma^{s-t} \rho_s^{\beta_k}(y), \quad \gamma \in [0, 1]. \quad (18)$$

When $\gamma = 1$, this recovers the exact sequence-level gradient estimator; in practice, we use $\gamma < 1$ to control the magnitude of the return on long generations. The proof is in Appendix B.3.

The practical β -OPSD loss is

$$\mathcal{L}_\beta(\theta; y) = \frac{1}{T} \sum_{t=1}^T \text{sg} \left(G_{t,\gamma}^{\beta_k}(y) \right) \log \pi_\theta(y_t | x, y_{<t}), \quad (19)$$

where $\text{sg}(\cdot)$ denotes stop-gradient. The interpolant logits and return-to-go weights are detached; the only gradient path is through the student log-probability. Thus, β -OPSD differs from vanilla OPD in two simple ways:

1. it replaces the fixed teacher target with a scheduled logit-interpolant target, and
2. it replaces local token weights with return-to-go weights for sequence-level credit assignment.

The first modification is the main conceptual contribution: it follows from the β -OPSD optimal policy. The second is a practical sequence-level estimator that makes the distillation loss faithful to the trajectory objective.

3. Related Work

The method above connects two themes that are often treated separately: how to choose a stable teacher target, and how to assign credit along an on-policy trajectory. We organize related work around this split. Prior work provides ingredients for smoother supervision, reference-regularized policy optimization, and return-based credit assignment; β -OPSD combines these ingredients into a single distillation procedure.

On-policy distillation and self-distillation. On-policy distillation trains the student on its own generated trajectories, using dense teacher feedback at each token. This avoids the exposure-bias mismatch of purely offline distillation, but introduces a new challenge: the teacher must provide reliable supervision on student-generated prefixes, which may be outside the teacher’s typical distribution (Li et al., 2026, Jang et al., 2026). Recent self-distillation methods further instantiate the teacher as the same model with access to privileged information, such as ground-truth solutions or feedback (Zhao et al., 2026, Liu et al., 2026, Yang et al., 2026, Shen et al., 2026, Zhang et al., 2026, Hübotter et al., 2026, He et al., 2026). β -OPSD belongs to this line of work, but differs by replacing the fixed teacher target with a scheduled student–teacher interpolant and by correcting the myopic local token-KL gradient with a Look-Ahead return.

RLHF and reference-regularized policy optimization. Our theoretical formulation is closely related to KL-regularized reinforcement learning and RLHF, where a policy is optimized to maximize reward while staying close to a reference model (Ziegler et al., 2019, Ouyang et al., 2022, Schulman et al., 2017). DPO shows that the optimal policy of a KL-regularized preference objective can be written in closed form, revealing a direct connection between rewards, reference policies, and optimal distributions. We use a similar perspective to reinterpret on-policy distillation: the teacher-to-reference log-probability ratio acts as a reward, and standard OPD is recovered as the $\beta = 1$ special case. This view leads directly to the geometric interpolant between the reference student and the privileged teacher.

Interpolation, curricula, and teacher-guided learning. Curriculum learning aims to make optimization smoother by gradually changing the difficulty or source of supervision (Bengio et al., 2009). Related ideas also appear in sequence prediction and imitation learning: scheduled sampling gradually moves training from teacher-forced inputs toward model-generated inputs (Bengio et al., 2015), while DAGger addresses distribution shift by querying expert supervision on learner-induced states (Ross et al., 2011). More recent work has explored student–teacher mixture reformulations to bridge the distributional gap in distillation (Jang et al., 2026), as well as teacher-guided or student–teacher cooperative sampling for improving training

rollouts (Xu et al., 2025, Shenfeld et al., 2023). Our logit-interpolant target can be viewed as a distributional curriculum: early targets remain close to the student, while later targets move toward the privileged teacher. Unlike heuristic interpolation or guidance used only for sampling, our interpolant is derived as the optimal policy of a reference-regularized distillation objective. This provides a principled target path from student to teacher and, when combined with return-to-go credit assignment, yields a stable and future-aware on-policy distillation algorithm.

Policy gradients and credit assignment. The return-to-go estimator used in β -OPSD is closely related to classical policy-gradient methods such as REINFORCE (Williams, 1992), where unbiased gradient estimation assigns each action its return-to-go rather than only its instantaneous reward. In OPD, the analogous per-step signal is the teacher–student log-ratio along a generated trajectory. Prior work has explored using local token-level signals for stable distillation (Li et al., 2026, Armandpour et al., 2026, Shen et al., 2026, Jang et al., 2026, Kim et al., 2026), and related derivations have also identified unbiased sequence-level gradient estimators for reverse-KL objectives (Agrawal et al., 2026, Liu et al., 2026). Future-aware credit assignment has also been studied in GRPO-style optimization (Schulman et al., 2015, Sutton et al., 1998). Our work builds on this perspective and shows that, for on-policy distillation, the standard local token-KL update is myopic because it ignores the future distributional mismatch induced by earlier tokens. β -OPSD applies return-to-go credit assignment directly to teacher-guided distillation: the exact undiscounted Look-Ahead estimator recovers an unbiased sequence-level KL gradient under student on-policy sampling, while discounted implementations provide a practical stability–variance trade-off for long generations.

4. Experiments

4.1. Experimental Setup

Models and Datasets. We evaluate β -OPSD on the instruct-tuned Qwen3 model family, including Qwen3-1.7B, Qwen3-4B, and Qwen3-8B. For training, we use the mathematical reasoning subset of OpenThoughts. Each training example contains a problem x and a ground-truth solution, which we use as privileged information c for the teacher branch. The student generates solutions conditioned only on the problem x , while the privileged teacher can condition on both x and c during training.

We evaluate on competition-level mathematical reasoning benchmarks, including AIME 2024, AIME 2025, and HMMT 2025. These benchmarks require multi-step reasoning and therefore test whether the method improves the quality of generated reasoning trajectories rather than only local token imitation.

Baselines. We compare against three classes of baselines. First, we include the base model without post-training. Second, we compare with supervised fine-tuning (SFT), which trains directly on ground-truth solutions with the standard next-token prediction objective. Third, we compare with vanilla on-policy self-distillation (OPSD), which distills directly toward the privileged teacher using a local token-KL objective. In our notation, vanilla OPSD uses the teacher endpoint as the target but does not use the scheduled interpolant curriculum or the return-to-go credit-assignment estimator. We also compare with GRPO-style reinforcement learning baselines, which optimize the student from sampled generations using outcome-level rewards.

Implementation Details. For the main β -OPSD experiments, we use a bounded linear path for the teacher weight rather than starting exactly at the student endpoint or ending exactly at the teacher endpoint. Let

$$s_k = \frac{k}{K-1}, \quad k = 0, \dots, K-1, \quad (20)$$

be the normalized training progress. We define the practical teacher weight

$$w_k = w_{\text{start}} + (w_{\text{end}} - w_{\text{start}}) s_k, \quad (21)$$

where $w_k = 1/\beta_k$. Thus, w_k is the teacher-side weight in the logit interpolant, while β_k is the corresponding regularization coefficient in the β -OPSD view.

At each update, we construct the interpolant target from a stop-gradient copy of the current student and the privileged teacher. For a prefix $h_t = (x, y_{<t})$, we use

$$\tilde{p}_{\beta_k}(\cdot \mid h_t, c) = \text{softmax}((1 - w_k)z_{\bar{\theta}}(\cdot \mid h_t) + w_k z_T(\cdot \mid h_t, c)), \quad (22)$$

where $\bar{\theta} = \text{sg}(\theta)$ denotes a stop-gradient copy of the current student. The token-level mismatch is

$$\rho_t^{\beta_k}(y) = \log \pi_{\theta}(y_t \mid x, y_{<t}) - \log \tilde{p}_{\beta_k}(y_t \mid x, y_{<t}, c). \quad (23)$$

We then compute the discounted return-to-go in Eq. 18 and the β -OPSD loss in Eq. 19

$$G_{t,\gamma}^{\beta_k}(y) = \sum_{s=t}^T \gamma^{s-t} \rho_s^{\beta_k}(y), \quad (24)$$

and optimize the practical β -OPSD loss

$$\mathcal{L}_{\beta\text{-OPSD}}(\theta; y) = \frac{1}{T} \sum_{t=1}^T \text{sg} \left(G_{t,\gamma}^{\beta_k}(y) \right) \log \pi_{\theta}(y_t \mid x, y_{<t}). \quad (25)$$

Unless otherwise specified, we use $w_{\text{start}} = 0.5$, $w_{\text{end}} = 0.8$, $K = 200$, and $\gamma = 0.99$. These choices keep early targets relatively close to the student while still exposing the model to stronger teacher guidance later in training. All experiments are conducted on NVIDIA RTX A6000 or NVIDIA H200 GPUs with LoRA (Hu et al., 2022). Additional training, generation, and evaluation details are provided in Appendix C.

4.2. Results on Math Reasoning

Table 1 reports avg@12 performance for Qwen3 models evaluated at the 100-step checkpoint from the default 200-step training schedule. Across all three model scales, β -OPSD improves the overall average over Vanilla OPSD and achieves the best average performance among all evaluated methods. The gains are most pronounced for Qwen3-1.7B, where β -OPSD improves over Vanilla OPSD by 9.16, 5.27, and 2.78 points on AIME 2024, AIME 2025, and HMMT 2025, respectively, yielding a 5.74-point improvement in average performance.

The larger-scale results show that the method remains competitive as model size increases. On Qwen3-4B and Qwen3-8B, β -OPSD improves the overall average over Vanilla OPSD by 1.76 and 1.66 points, respectively. While it slightly underperforms Vanilla OPSD on some individual benchmarks, such as AIME 2024 for

Table 1: Avg@12 results for Qwen3 models evaluated at the 100-step checkpoint from the default 200-step training schedule. Gains denote absolute differences of β -OPSD relative to Vanilla OPSD within the same model scale. For β -OPSD, we report results under linear schedules along the target path.

Method	AIME 2024	AIME 2025	HMMT 2025	Average
<i>Qwen3-8B</i>				
Base model	73.06	67.78	29.72	56.85
Vanilla OPSD	75.83	66.67	33.06	58.52
SFT	72.78	64.72	30.83	56.11
GRPO	75.83	68.61	31.11	58.52
β-OPSD (ours)	78.33 (+2.50)	70.28 (+3.61)	31.94 (-1.12)	60.18 (+1.66)
<i>Qwen3-4B</i>				
Base model	77.78	64.17	29.44	57.13
Vanilla OPSD	75.00	65.00	28.33	56.11
SFT	23.61	19.72	10.56	17.96
GRPO	76.39	64.17	30.28	56.95
β-OPSD (ours)	73.89 (-1.11)	69.17 (+4.17)	30.56 (+2.23)	57.87 (+1.76)
<i>Qwen3-1.7B</i>				
Base model	50.00	37.22	14.44	33.89
Vanilla OPSD	44.17	35.56	13.33	31.02
SFT	37.50	29.17	12.22	26.30
GRPO	48.33	38.61	15.00	33.98
β-OPSD (ours)	53.33 (+9.16)	40.83 (+5.27)	16.11 (+2.78)	36.76 (+5.74)

Qwen3-4B and HMMT 2025 for Qwen3-8B, it gives the strongest overall performance across the benchmark suite.

Compared with other post-training baselines, β -OPSD consistently outperforms SFT across all model scales and achieves stronger performance than GRPO. Overall, these results support our central claim: replacing direct teacher imitation with scheduled logit-interpolant targets provides a smoother and more effective distillation objective. The advantage is especially clear for smaller models, where the student-teacher distributional gap is larger, and remains competitive or stronger at larger scales.

4.3. Ablation Studies

We conduct controlled ablations to isolate the contribution of each component of β -OPSD. Unless otherwise specified, all variants use the same student, privileged teacher, training data, optimization hyperparameters, training budget, and evaluation protocol as the main experiments. We vary only the target distribution, credit-assignment estimator, interpolant reference, or interpolation schedule under study. All ablations are conducted with Qwen3-1.7B and report avg@12 on AIME 2024, AIME 2025, and HMMT 2025.

4.3.1. Effect of Logit Interpolants

We first study whether the improvement of β -OPSD comes from the logit-interpolant target or solely from the return-to-go credit-assignment estimator. To isolate the effect of the target distribution, we apply the same return-to-go loss to two different distillation targets.

Algorithm 1 β -OPSD: Smooth Target Distillation Toward Logit Interpolants

Require: Student π_θ , privileged teacher p_T , dataset $\mathcal{D}$, training budget K , schedule endpoints $w_{\text{start}}, w_{\text{end}}$, discount factor γ

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1: for  $k = 0, \dots, K - 1$  do
2:   Sample a minibatch  $(x, c) \sim \mathcal{D}$ 
3:   Sample on-policy responses  $y = (y_1, \dots, y_T) \sim \pi_\theta(\cdot | x)$ 
4:   Set normalized progress  $s_k = k / (K - 1)$ 
5:   Set teacher weight  $w_k$  by Eq. 21 and  $\beta_k = 1 / w_k$ 
6:   Set  $\bar{\theta} \leftarrow \text{sg}(\theta)$ 
7:   Construct interpolant targets  $\tilde{p}_{\beta_k}(\cdot | h_t, c)$  along the sampled prefixes using Eq. 22
8:   Compute token mismatches  $\rho_t^{\beta_k}(y)$  using Eq. 23
9:   Compute discounted return-to-go weights  $G_{t,\gamma}^{\beta_k}(y)$  using Eq. 24
10:  Minimize  $\mathcal{L}_{\beta\text{-OPSD}}(\theta; y)$  in Eq. 25 and update  $\theta$ 
11: end for
  
```

Table 2: Effect of the distillation target. Both variants use the same return-to-go estimator. We report avg@12 (%); the best result in each column is bolded. For the β -OPSD target, we report the result with $w_{\text{start}} = 0.5$ and $w_{\text{end}} = 0.8$.

Method	AIME 2024	AIME 2025	HMMT 2025
Teacher + Return-to-go	47.30	35.53	14.44
β-OPSD target	53.33	40.83	16.11

The first variant, *Teacher + Return-to-go*, distills directly toward the privileged teacher. Its token-level log-ratio is

$$\rho_t^{\text{teach}}(y) = \log \pi_\theta(y_t | x, y_{<t}) - \log p_T(y_t | x, y_{<t}, c), \quad (26)$$

with return-to-go $G_{t,\gamma}^{\text{teach}}(y) = \sum_{s=t}^T \gamma^{s-t} \rho_s^{\text{teach}}(y)$.

The second variant uses the scheduled β -OPSD target. Its token-level log-ratio is

$$\rho_t^{\beta_k}(y) = \log \pi_\theta(y_t | x, y_{<t}) - \log \tilde{p}_{\beta_k}(y_t | x, y_{<t}, c), \quad (27)$$

with return-to-go $G_{t,\gamma}^{\beta_k}(y) = \sum_{s=t}^T \gamma^{s-t} \rho_s^{\beta_k}(y)$.

Both variants use student on-policy trajectories and identical return-to-go credit assignment. Therefore, this comparison isolates the benefit of replacing the fixed teacher target with the scheduled logit-interpolant target.

As shown in Table 2, the β -OPSD target consistently outperforms direct teacher distillation, improving avg@12 by 6.03 points on AIME 2024, 5.30 points on AIME 2025, and 1.67 points on HMMT 2025. These gains highlight the importance of a smooth transition in the target objective: rather than abruptly projecting the student onto the teacher, the interpolant target reduces the student–target gap and provides a more stable learning signal.

4.3.2. Effect of Return-to-go Credit Assignment

We next isolate the contribution of return-to-go credit assignment. Both variants use the same logit-interpolant target $\tilde{p}_{\beta_k}$ and the same student-generated trajectories, but differ in how the token-level signal is assigned.

Table 3: Effect of return-to-go credit assignment. Both variants use the same logit-interpolant target with fixed schedules $w_k = \frac{1}{\beta_k} = 0.5$. We report avg@12 (%).

Method	AIME 2024	AIME 2025	HMMT 2025
Local Token Gradient	49.44	32.78	13.06
Return-to-go Gradient	50.56	38.33	16.67

The *Local Token Gradient* variant uses only the instantaneous log-ratio:

$$\hat{g}_{\text{local}}(y) = \sum_{t=1}^T \rho_t^{\beta_k}(y) \nabla_{\theta} \log \pi_{\theta}(y_t \mid x, y_{<t}). \quad (28)$$

In contrast, the *Return-to-go* variant uses the cumulative future mismatch:

$$\hat{g}_{\text{RTG}}(y) = \sum_{t=1}^T G_{t,\gamma}^{\beta_k}(y) \nabla_{\theta} \log \pi_{\theta}(y_t \mid x, y_{<t}). \quad (29)$$

As shown in Table 3, return-to-go credit assignment consistently outperforms the local token-gradient objective. It improves avg@12 by 1.12 points on AIME 2024, 5.55 points on AIME 2025, and 3.61 points on HMMT 2025. Since the two variants share the same target and sampling distribution, these gains isolate the benefit of assigning earlier tokens credit for the future KL mismatch they induce.

4.3.3. Effect of Interpolant References

We ablate how the two endpoints of the logit interpolant are constructed. Let θ_0 denote the initial student parameters, let θ_k denote the student parameters at training step k , and let $\bar{\theta}_k = \text{sg}(\theta_k)$ denote a stop-gradient copy of the current student. In our self-distillation setting, the teacher branch uses the same model architecture but additionally conditions on privileged information c .

We compare three reference choices. First, *fixed student + fixed teacher* constructs both endpoints from the initial model:

$$\tilde{p}_w^{\text{FF}}(\cdot \mid h_t, c) = \text{softmax}((1-w)z_{\theta_0}(\cdot \mid h_t) + wz_{\theta_0}(\cdot \mid h_t, c)). \quad (30)$$

Second, *dynamic student + dynamic teacher* refreshes both endpoints from the current model:

$$\tilde{p}_w^{\text{DD}}(\cdot \mid h_t, c) = \text{softmax}((1-w)z_{\bar{\theta}_k}(\cdot \mid h_t) + wz_{\bar{\theta}_k}(\cdot \mid h_t, c)). \quad (31)$$

Third, *dynamic student + fixed teacher*, our default construction, uses the current student as the student-side endpoint and a fixed privileged teacher as the teacher-side endpoint:

$$\tilde{p}_w^{\text{DF}}(\cdot \mid h_t, c) = \text{softmax}((1-w)z_{\bar{\theta}_k}(\cdot \mid h_t) + wz_{\theta_0}(\cdot \mid h_t, c)). \quad (32)$$

To isolate the effect of the interpolant references, we evaluate all three constructions using the same return-to-go estimator and identical training settings. We keep the teacher weight fixed at $w = 0.5$, so that the two endpoints contribute equally to the target distribution. As shown in Figure 2, the dynamic-student and fixed-teacher construction performs best across all three benchmarks. These results suggest that the interpolant benefits from tracking the evolving student distribution while retaining a stable privileged teacher as its supervisory anchor.

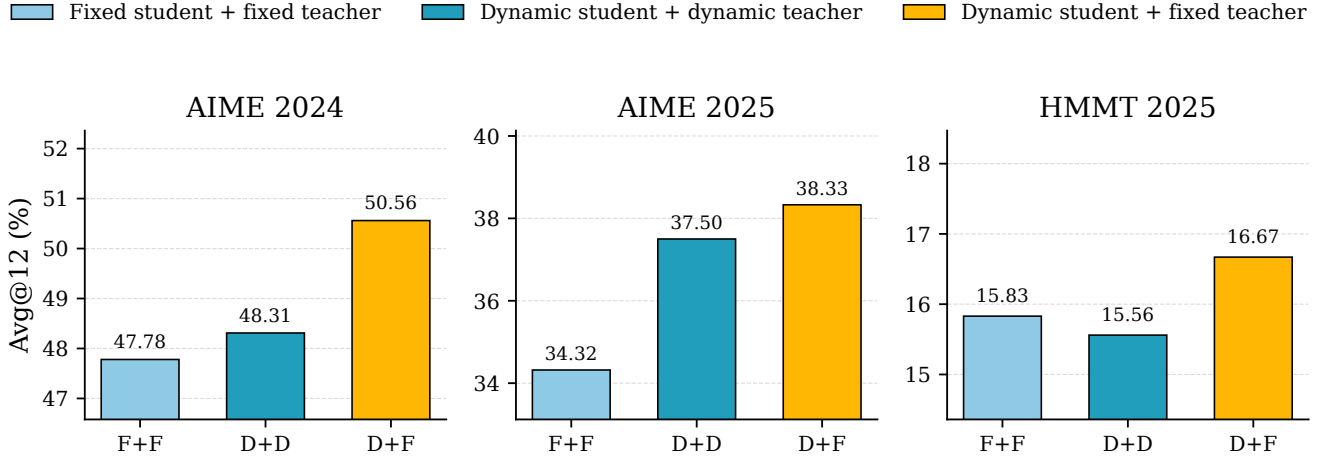


Figure 2: Effect of interpolant reference choices. All variants use the same return-to-go estimator and fixed teacher weight $w_k = 1/\beta_k = 0.5$. We report avg@12. Each subplot uses a benchmark-specific y-axis range to highlight relative differences. The dynamic student + fixed teacher construction performs best across all benchmarks.

Table 4: Effect of interpolation schedules. The notation $a \rightarrow b$ indicates that the teacher-weight endpoints $w_{\text{start}} = a$ and $w_{\text{end}} = b$ are using the schedule in Eq 21. We report avg@12 (%).

Schedule	AIME 2024	AIME 2025	HMMT 2025
Fixed 0.5	50.56	38.33	16.67
Linear 0.2 $\rightarrow$ 0.8	51.39	37.50	16.39
Linear 0.8 $\rightarrow$ 0.2	51.94	38.06	15.83
Linear 0.5 $\rightarrow$ 0.8	53.33	40.83	16.11
Linear 0.8 $\rightarrow$ 0.5	54.72	38.61	15.83

4.3.4. Effect of the Interpolation Schedule

Finally, we study how the interpolation schedule affects performance. We focus on linear schedules for the teacher weight $w_k = 1/\beta_k$, varying the start and end values in Eq. 21. We also include a fixed $w_k = 0.5$ baseline.

As shown in Table 4, the schedule influences performance across benchmarks. The $0.5 \rightarrow 0.8$ schedule gives the strongest overall result among the schedules considered here, while other schedules can be better on individual benchmarks. This suggests that the strength and timing of teacher guidance are important practical choices. Beyond the linear schedules studied here, nonlinear, piecewise, or performance-adaptive schedules may further improve performance. Determining whether an optimal schedule exists, and how it depends on model scale, training budget, and task distribution, is an interesting direction for future work.

Remark 4.1. In addition to β -OPSD, which changes the distillation target through scheduled logit interpolants, one can also construct guided rollouts by sampling from a mixture of the student and the privileged teacher. This variant changes the trajectory distribution rather than only the target distribution, and therefore requires importance correction. We provide details and results in Appendix D.

5. Conclusion

We introduced β -OPSD, a framework built on the view that on-policy distillation is trajectory-level policy optimization rather than merely local token imitation. Under this view, standard sequence-level OPSD is the $\beta = 1$ teacher endpoint of a reference-regularized RL family. In the conservative regime, the optimal policies form a geometric path from a reference student to the privileged teacher, motivating tractable token-level logit interpolants instead of direct teacher matching throughout training.

β -OPSD further replaces the myopic local token-KL update with a return-to-go credit assignment that accounts for future distributional mismatch and provides an unbiased gradient estimator for the sequence-level objective. Experiments on competition-level mathematical reasoning benchmarks show consistent improvements over vanilla OPSD, SFT, and GRPO methods. Our ablations confirm the importance of both interpolant targets and future-aware credit assignment. These results suggest that smooth target paths and trajectory-level credit assignment are effective principles for on-policy distillation. Exploring adaptive or theoretically optimized interpolation schedules, lower-variance return-to-go estimators, and applications beyond mathematical reasoning are promising directions for future work.

6. Acknowledgment

Xu, Liu and Huang are supported by DARPA HR001124S0029-AIQ-FP-019, National Science Foundation TRAILS Institute (2229885). Private support was provided by Open Philanthropy and Apple. The Authors acknowledge the National Artificial Intelligence Research Resource (NAIRR) Pilot for contributing to this research result.

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A. Notation Reference

Table 5: Summary of notation. For readability, we omit conditioning variables such as (x, c) when they are clear from context. Throughout the paper, π_θ denotes the student policy, π_{ref} denotes the reference policy, and p_T denotes the privileged teacher.

Notation	Description
x	Input prompt.
c	Privileged context available only during training.
$y = (y_1, \dots, y_T)$	Generated reasoning trajectory.
y_t	Token at decoding step t .
$y_{<t}$	Prefix before token t .
$h_t = (x, y_{<t})$	Decoding history or prefix state at step t .
$\pi_\theta(y \mid x)$	Student policy parameterized by θ .
$p_T(y \mid x, c)$	Privileged teacher policy conditioned on prompt x and context c .
$\pi_{\text{ref}}(y \mid x)$	Reference policy, e.g., the initial student or a stop-gradient copy.
$\tilde{\theta}$	Stop-gradient copy of the current student parameters.
$R(y; x, c)$	Trajectory-level reward in the KL-regularized RL formulation.
$\mathcal{J}_\beta(\theta)$	β -OPSD objective, obtained by using the teacher-to-reference log-ratio reward.
β	KL regularization coefficient in β -OPSD.
β_k	Regularization coefficient used at training step k .
$\pi_\beta^*(y \mid x, c)$	Closed-form optimal policy of the β -OPSD objective.
$z_{\text{ref}}(\cdot \mid h_t)$	Reference/student logits at prefix h_t .
$z_T(\cdot \mid h_t, c)$	Privileged teacher logits at prefix h_t .
$\tilde{p}_{\beta_k}(\cdot \mid h_t, c)$	Local logit-interpolant target at step k .
$\tilde{p}_{\beta_k}(y \mid x, c)$	Autoregressive sequence-level logit-interpolant target.
$s_k = \frac{k}{K-1}$	Normalized training progress.
w_k	Implementation notation for the teacher weight.
$\rho_t^{\beta_k}(y)$	Token-level log-ratio between the student and interpolant target.
$G_{t,\gamma}^{\beta_k}(y)$	Discounted return-to-go from token t using future interpolant log-ratios.
γ	Discount factor for the return-to-go.
$\text{sg}(\cdot)$	Stop-gradient operator.
$D_{\text{KL}}(\cdot \parallel \cdot)$	Kullback–Leibler divergence.
$\mathbb{E}_{y \sim \pi_\theta}$	Expectation over trajectories sampled from the student policy.

B. Theoretical Proofs

B.1. Proof of Proposition 2.1

Proof. Fix a prompt x and privileged context c . For readability, we suppress the conditioning on (x, c) when it is unambiguous. Expanding Eq. 3, we have

$$\mathcal{J}_\beta(\pi_\theta) = \mathbb{E}_{y \sim \pi_\theta} \left[\log \frac{p_T(y)}{\pi_{\text{ref}}(y)} \right] - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \quad (33)$$

$$= \mathbb{E}_{y \sim \pi_\theta} [\log p_T(y) - \log \pi_{\text{ref}}(y)] - \beta \mathbb{E}_{y \sim \pi_\theta} [\log \pi_\theta(y) - \log \pi_{\text{ref}}(y)] \quad (34)$$

$$= \mathbb{E}_{y \sim \pi_\theta} [\log p_T(y) - \beta \log \pi_\theta(y) + (\beta - 1) \log \pi_{\text{ref}}(y)]. \quad (35)$$

On the other hand,

$$-D_{\text{KL}}(\pi_\theta \| p_T) - (\beta - 1) D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \quad (36)$$

$$= -\mathbb{E}_{y \sim \pi_\theta} [\log \pi_\theta(y) - \log p_T(y)] - (\beta - 1) \mathbb{E}_{y \sim \pi_\theta} [\log \pi_\theta(y) - \log \pi_{\text{ref}}(y)] \quad (37)$$

$$= \mathbb{E}_{y \sim \pi_\theta} [\log p_T(y) - \beta \log \pi_\theta(y) + (\beta - 1) \log \pi_{\text{ref}}(y)]. \quad (38)$$

Comparing this expression with Eq. 35, we obtain the equivalent decomposition

$$\mathcal{J}_\beta(\pi_\theta) = -D_{\text{KL}}(\pi_\theta(\cdot | x) \| p_T(\cdot | x, c)) - (\beta - 1) D_{\text{KL}}(\pi_\theta(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)). \quad (39)$$

Setting $\beta = 1$ cancels the reference-regularization term:

$$\mathcal{J}_1(\pi_\theta) = -D_{\text{KL}}(\pi_\theta(\cdot | x) \| p_T(\cdot | x, c)). \quad (40)$$

Therefore,

$$\max_{\theta} \mathcal{J}_1(\pi_\theta) \iff \min_{\theta} D_{\text{KL}}(\pi_\theta(\cdot | x) \| p_T(\cdot | x, c)), \quad (41)$$

which is exactly the standard sequence-level OPSD objective in Eq. 4. This proves the proposition. $\square$

B.2. Proof of Proposition 2.2

Proof. The β -OPSD objective:

$$\mathcal{J}_\beta(\pi) = \mathbb{E}_{y \sim \pi(\cdot | x)} \left[\log \frac{p_T(y | x, c)}{\pi_{\text{ref}}(y | x)} \right] - \beta D_{\text{KL}}(\pi(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)). \quad (42)$$

Expanding the KL term gives

$$\mathcal{J}_\beta(\pi) = \mathbb{E}_{y \sim \pi} [\log p_T(y) - \log \pi_{\text{ref}}(y)] - \beta \mathbb{E}_{y \sim \pi} [\log \pi(y) - \log \pi_{\text{ref}}(y)] \quad (43)$$

$$= \mathbb{E}_{y \sim \pi} [\log p_T(y) - \beta \log \pi(y) + (\beta - 1) \log \pi_{\text{ref}}(y)]. \quad (44)$$

Now define

$$\pi_\beta^*(y | x, c) = \frac{\pi_{\text{ref}}(y | x)^{1-\frac{1}{\beta}} p_T(y | x, c)^{\frac{1}{\beta}}}{Z_\beta(x, c)}, \quad (45)$$

where

$$Z_\beta(x, c) = \sum_{y'} \pi_{\text{ref}}(y' | x)^{1-\frac{1}{\beta}} p_T(y' | x, c)^{\frac{1}{\beta}}. \quad (46)$$

Taking logs, we have

$$\log \pi_\beta^*(y | x, c) = \left(1 - \frac{1}{\beta}\right) \log \pi_{\text{ref}}(y | x) + \frac{1}{\beta} \log p_T(y | x, c) - \log Z_\beta(x, c). \quad (47)$$

Multiplying both sides by β gives

$$\beta \log \pi_\beta^*(y | x, c) = (\beta - 1) \log \pi_{\text{ref}}(y | x) + \log p_T(y | x, c) - \beta \log Z_\beta(x, c). \quad (48)$$

Substituting Eq. 48 into Eq. 44, we obtain

$$\mathcal{J}_\beta(\pi) = \mathbb{E}_{y \sim \pi} \left[\beta \log \pi_\beta^*(y | x, c) + \beta \log Z_\beta(x, c) - \beta \log \pi(y | x) \right] \quad (49)$$

$$= -\beta \mathbb{E}_{y \sim \pi} \left[\log \frac{\pi(y | x)}{\pi_\beta^*(y | x, c)} \right] + \beta \log Z_\beta(x, c) \quad (50)$$

$$= -\beta D_{\text{KL}} \left(\pi(\cdot | x) \parallel \pi_\beta^*(\cdot | x, c) \right) + \beta \log Z_\beta(x, c). \quad (51)$$

The second term is independent of π , while the KL term is nonnegative and is minimized to zero if and only if

$$\pi(\cdot | x) = \pi_\beta^*(\cdot | x, c). \quad (52)$$

Therefore, π_β^* is the maximizer of Eq. 3. This proves the proposition. $\square$

B.3. Proof of the Return-to-Go Gradient Estimator

Proof. We prove that when $\gamma = 1$, the return-to-go estimator in Eq. 18 gives an unbiased estimator of the sequence-level reverse-KL gradient.

Proposition B.1 (Unbiased sequence-level gradient estimator). *Fix a training step k and treat the interpolant target $\tilde{p}_{\beta_k}(\cdot | x, c)$ as fixed during the current gradient update. Define*

$$\mathcal{L}_{\beta_k}(\theta) = D_{\text{KL}} \left(\pi_\theta(\cdot | x) \parallel \tilde{p}_{\beta_k}(\cdot | x, c) \right), \quad (53)$$

and let

$$\rho_t^{\beta_k}(y) = \log \pi_\theta(y_t | x, y_{<t}) - \log \tilde{p}_{\beta_k}(y_t | x, y_{<t}, c). \quad (54)$$

For $\gamma = 1$, define

$$G_t^{\beta_k}(y) = \sum_{s=t}^T \rho_s^{\beta_k}(y). \quad (55)$$

Then

$$\widehat{g}_{\text{RTG}}(y) = \sum_{t=1}^T G_t^{\beta_k}(y) \nabla_\theta \log \pi_\theta(y_t | x, y_{<t}) \quad (56)$$

satisfies

$$\mathbb{E}_{y \sim \pi_\theta(\cdot | x)} [\widehat{g}_{\text{RTG}}(y)] = \nabla_\theta \mathcal{L}_{\beta_k}(\theta). \quad (57)$$

Proof. For readability, fix x, c, β_k and suppress them when clear. By the autoregressive factorization,

$$\pi_\theta(y \mid x) = \prod_{t=1}^T \pi_\theta(y_t \mid x, y_{<t}), \quad \tilde{p}_{\beta_k}(y \mid x, c) = \prod_{t=1}^T \tilde{p}_{\beta_k}(y_t \mid x, y_{<t}, c). \quad (58)$$

Therefore,

$$\mathcal{L}_{\beta_k}(\theta) = \mathbb{E}_{y \sim \pi_\theta(\cdot \mid x)} \left[\sum_{t=1}^T \rho_t^{\beta_k}(y) \right]. \quad (59)$$

Let

$$R^{\beta_k}(y) := \sum_{t=1}^T \rho_t^{\beta_k}(y), \quad g_t(y) := \nabla_\theta \log \pi_\theta(y_t \mid x, y_{<t}). \quad (60)$$

Then

$$\mathcal{L}_{\beta_k}(\theta) = \mathbb{E}_{y \sim \pi_\theta} [R^{\beta_k}(y)]. \quad (61)$$

Differentiating with the score-function identity gives

$$\nabla_\theta \mathcal{L}_{\beta_k}(\theta) = \mathbb{E}_{y \sim \pi_\theta} [R^{\beta_k}(y) \nabla_\theta \log \pi_\theta(y \mid x)] + \mathbb{E}_{y \sim \pi_\theta} [\nabla_\theta R^{\beta_k}(y)]. \quad (62)$$

Since

$$\nabla_\theta \log \pi_\theta(y \mid x) = \sum_{t=1}^T g_t(y), \quad (63)$$

and the interpolant target $\tilde{p}_{\beta_k}$ is fixed during the update,

$$\nabla_\theta R^{\beta_k}(y) = \sum_{t=1}^T g_t(y). \quad (64)$$

The expected score is zero:

$$\mathbb{E}_{y \sim \pi_\theta} \left[\sum_{t=1}^T g_t(y) \right] = 0. \quad (65)$$

Thus,

$$\nabla_\theta \mathcal{L}_{\beta_k}(\theta) = \mathbb{E}_{y \sim \pi_\theta} \left[R^{\beta_k}(y) \sum_{t=1}^T g_t(y) \right] \quad (66)$$

$$= \mathbb{E}_{y \sim \pi_\theta} \left[\sum_{t=1}^T g_t(y) \sum_{s=1}^T \rho_s^{\beta_k}(y) \right]. \quad (67)$$

We now remove the past terms by causality. For $s < t$, the quantity $\rho_s^{\beta_k}(y)$ depends only on earlier tokens and is fixed after conditioning on the prefix $(x, y_{<t})$. Meanwhile,

$$\mathbb{E}_{y_t \sim \pi_\theta(\cdot \mid x, y_{<t})} [g_t(y) \mid x, y_{<t}] = \sum_{v \in \mathcal{V}} \pi_\theta(v \mid x, y_{<t}) \nabla_\theta \log \pi_\theta(v \mid x, y_{<t}) \quad (68)$$

$$= \sum_{v \in \mathcal{V}} \nabla_\theta \pi_\theta(v \mid x, y_{<t}) \quad (69)$$

$$= \nabla_\theta \sum_{v \in \mathcal{V}} \pi_\theta(v \mid x, y_{<t}) \quad (70)$$

$$= 0. \quad (71)$$

Therefore,

$$\mathbb{E}_{y \sim \pi_\theta} \left[g_t(y) \sum_{s < t} \rho_s^{\beta_k}(y) \right] = 0. \quad (72)$$

Applying this to Eq. 67, only the current and future terms remain:

$$\nabla_\theta \mathcal{L}_{\beta_k}(\theta) = \mathbb{E}_{y \sim \pi_\theta} \left[\sum_{t=1}^T g_t(y) \sum_{s=t}^T \rho_s^{\beta_k}(y) \right] \quad (73)$$

$$= \mathbb{E}_{y \sim \pi_\theta} \left[\sum_{t=1}^T G_t^{\beta_k}(y) g_t(y) \right]. \quad (74)$$

This proves that the return-to-go estimator is unbiased for the sequence-level gradient. $\square$

Finally, the practical surrogate in Eq. 19 with $\gamma = 1$ has gradient

$$\nabla_\theta \mathcal{L}_{\beta\text{-OPSD}}(\theta; y) = \frac{1}{T} \sum_{t=1}^T G_t^{\beta_k}(y) \nabla_\theta \log \pi_\theta(y_t \mid x, y_{<t}), \quad (75)$$

where the return weights are detached. Hence, its expectation equals $\frac{1}{T} \nabla_\theta \mathcal{L}_{\beta_k}(\theta)$. The factor $1/T$ is only a normalization constant. For $\gamma < 1$, the same loss becomes a discounted practical approximation that reduces the magnitude and variance of long-horizon returns, but the exact unbiasedness statement corresponds to $\gamma = 1$. $\square$

C. Experimental Details

C.1. Training Details

Shared training configuration. Vanilla OPSD and β -OPSD use the same model, optimization settings, rollout configuration, and computational budget. We train Qwen3-1.7B on four NVIDIA RTX A6000 GPUs using four-way data parallelism with `accelerate`. Each training rank runs a colocated single-GPU vLLM instance for on-policy generation. Training is run for a maximum of 200 optimization steps with an eight-hour wall-time limit.

Optimization and parameter-efficient tuning. We use a learning rate of 5×10^{-6} and clip the gradient norm at 0.1. The per-device batch size is 1, with 8 gradient-accumulation steps, giving an effective batch size of

$$4 \text{ GPUs} \times 1 \text{ sample per GPU} \times 8 \text{ accumulation steps} = 32. \quad (76)$$

Gradient checkpointing is enabled. All models are trained in bfloat16 precision using FlashAttention-2. We use LoRA with rank $r = 64$ and scaling parameter $\alpha_{\text{LoRA}} = 128$. LoRA adapters are applied to `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`.

On-policy generation. Training trajectories are generated with a maximum sequence length of 20,000 tokens and a maximum completion length of 1,024 tokens. We sample with temperature 1.1, $\text{top-}p = 0.95$, and $\text{top-}k = 20$. Generation uses vLLM in colocated mode with tensor-parallel size 1 and GPU-memory utilization set to 0.5. Unless otherwise stated, these settings are shared by Vanilla OPSD and β -OPSD, ensuring that their comparison isolates the effect of the distillation objective and Look-Ahead credit assignment.

C.2. Evaluation Details

Evaluation pipeline. We evaluate the Qwen3-1.7B, Qwen3-4B and Qwen3-8B checkpoints using their corresponding LoRA adapters loaded on top of the original Qwen3 base models. We evaluate on AIME 2024, AIME 2025, and HMMT 2025.

Decoding configuration. We use stochastic decoding with thinking mode enabled. For each problem, we generate $k = 12$ independent candidate solutions using temperature 0.6, top- p sampling with $p = 0.95$, and top- $k = 50$. The evaluation batch size is 32. We set `max_new_tokens=0`, corresponding to no separate completion-length cap in the evaluation script, and use the default maximum model length of 40,960 tokens in thinking mode. Unless explicitly changed for an ablation, all reported results use this same decoding configuration.

Scoring and metrics. We report *pass@12*, the fraction of problems for which at least one of the 12 sampled solutions is correct, and *avg@12*, the average correctness across all 12 sampled solutions. The evaluation code returns both metrics as raw fractions; throughout the paper, we multiply them by 100 and report percentage points.

Inference configuration. Evaluation uses tensor-parallel size 1, GPU memory utilization 0.9, and a single evaluation repeat. LoRA checkpoints are evaluated without modifying the underlying base model weights. Checkpoint sweeps evaluate training steps $\{50, 75, 100, 200\}$, with one Slurm evaluation job submitted for each checkpoint–dataset pair.

D. Sampling from Logit Interpolants

In addition to student on-policy sampling, we evaluate a guided sampling variant that constructs trajectories from a mixture of the student and privileged teacher. For proposal-mixture weight $\eta \in [0, 1]$, we define the token-level proposal distribution

$$m_{\bar{\theta}, \eta}(\cdot \mid h_t, c) = (1 - \eta) \pi_{\bar{\theta}}(\cdot \mid h_t) + \eta p_T(\cdot \mid h_t, c), \quad (77)$$

where $\pi_{\bar{\theta}}$ is a stop-gradient copy of the current student. Thus, $\eta = 0$ recovers student on-policy sampling, whereas $\eta = 1$ samples entirely from the privileged teacher. The resulting autoregressive proposal is

$$m_{\bar{\theta}, \eta}(y \mid x, c) = \prod_{t=1}^T m_{\bar{\theta}, \eta}(y_t \mid x, y_{<t}, c). \quad (78)$$

Because the objective in Eq. 4 is defined under the student trajectory distribution π_{θ} , sampling from $m_{\bar{\theta}, \eta}$ introduces a distribution mismatch. We correct this mismatch using per-decision importance sampling. For a trajectory $y \sim m_{\bar{\theta}, \eta}(\cdot \mid x, c)$, we define

$$w_t(y) = \frac{\pi_{\theta}(y_t \mid x, y_{<t})}{m_{\bar{\theta}, \eta}(y_t \mid x, y_{<t}, c)}, \quad W_t(y) = \prod_{i=1}^t w_i(y). \quad (79)$$

The importance-weighted Look-Ahead return is

$$G_{t,\gamma}^{\eta,\text{mix}}(y) = \sum_{s=t}^T \gamma^{s-t} W_s(y) \rho_s^\tau(y), \quad (80)$$

where ρ_s^τ measures the student mismatch with the logit-interpolant target. We optimize the corresponding surrogate

$$\mathcal{L}_{\text{mix-}\beta\text{-OPSD}}(\theta; y) = \frac{1}{T} \sum_{t=1}^T \text{sg} \left(G_{t,\gamma}^{\tau,\text{mix}}(y) \right) \log \pi_\theta(y_t \mid x, y_{<t}). \quad (81)$$

We note that this variant is inherently off-policy and requires combining the student and teacher distributions at every decoding step. At the time of writing, such dual-model sampling is **not** directly supported by standard vLLM decoding pipelines, resulting in additional (nearly double) implementation complexity and generation overhead. Although mixed student–teacher sampling yields comparable improvements over vanilla OPSD, its reduced sampling efficiency makes it less attractive in practice.

Results We evaluate fixed proposal-mixture weights $\eta \in \{0.2, 0.5, 0.8\}$ and linear schedules that either increase or decrease η during training. Here, η denotes the teacher weight in Eq. 77, while $1 - \eta$ denotes the student weight. All mixed-sampling variants use β -OPSD with discount factor $\gamma = 0.99$ and per-decision importance sampling (IS).

Table 6 shows that mixed student–teacher sampling can produce improvements comparable to those of the main on-policy β -OPSD method. At 200 steps, every evaluated schedule improves avg@12 over Vanilla OPSD on both AIME benchmarks. Across all checkpoints, the best observed gains over Vanilla OPSD are 8.06 points on AIME 2024, 5.55 points on AIME 2025, and 3.06 points on HMMT 2025. However, no single proposal schedule dominates across all benchmarks, indicating that the off-policy variant is sensitive to both the mixture coefficient and the training checkpoint. Given its additional dual-model decoding cost and the variance introduced by importance sampling, we use student on-policy sampling as the default implementation of β -OPSD.

Table 6: Mixed student–teacher sampling results. All non-baseline variants use Sampling from Logit Interpolants with discount factor $\gamma = 0.99$ and importance sampling (IS). Here, η denotes the teacher weight in the proposal distribution. We report pass@12 and avg@12 in percentage points.

Proposal schedule	Step	AIME 2024		AIME 2025		HMMT 2025	
		pass@12	avg@12	pass@12	avg@12	pass@12	avg@12
Vanilla OPSD	200	76.67	46.11	63.33	35.56	43.33	15.83
<i>Fixed proposal: $\eta = 0.2$</i>							
Mixed Sampling	50	76.67	50.83	60.00	37.50	40.00	15.83
Mixed Sampling	75	80.00	50.00	66.67	38.06	40.00	14.44
Mixed Sampling	100	76.67	52.78	53.33	35.83	33.33	13.61
Mixed Sampling	200	73.33	51.94	63.33	37.22	43.33	16.11
<i>Fixed proposal: $\eta = 0.5$</i>							
Mixed Sampling	50	80.00	50.83	66.67	36.39	43.33	16.11
Mixed Sampling	75	80.00	52.22	66.67	37.78	46.67	16.11
Mixed Sampling	100	76.67	54.17	63.33	39.44	46.67	17.22
Mixed Sampling	200	76.67	53.33	63.33	38.06	36.67	15.83
<i>Fixed proposal: $\eta = 0.8$</i>							
Mixed Sampling	50	80.00	50.83	60.00	36.94	43.33	15.83
Mixed Sampling	75	80.00	51.39	70.00	41.11	46.67	16.11
Mixed Sampling	100	73.33	51.94	60.00	36.11	40.00	15.83
Mixed Sampling	200	73.33	51.39	66.67	38.61	36.67	13.89
<i>Linear proposal schedule: $\eta : 0.2 \rightarrow 0.8$</i>							
Mixed Sampling	50	80.00	50.83	70.00	38.06	36.67	13.61
Mixed Sampling	75	76.67	50.56	70.00	36.39	53.33	18.06
Mixed Sampling	100	73.33	51.94	66.67	36.67	36.67	16.39
Mixed Sampling	200	76.67	50.56	60.00	41.11	43.33	16.94
<i>Linear proposal schedule: $\eta : 0.8 \rightarrow 0.2$</i>							
Mixed Sampling	50	76.67	50.56	76.67	36.39	30.00	13.61
Mixed Sampling	75	73.33	47.78	73.33	36.39	50.00	18.89
Mixed Sampling	100	80.00	50.56	60.00	38.61	36.67	15.83
Mixed Sampling	200	76.67	52.78	60.00	39.72	36.67	15.56